

activations were correlated with the agent’s x or y coordinates. We found 448 parameters that were significantly correlated with the agent’s x position across all SIR types, predominantly in intermediate and deeper layers. Shuffling the parameter indices between SIR types markedly reduced the number of common significant units (compare black and gray curves in the bottom second panel of Fig. 4, Wilcoxon signed-rank test, $W = 7$, $p < .001$). In contrast, units encoding the agent’s y coordinate regardless of SIR type were less common but still existed (258 in total) and tended to be concentrated in the most superficial layers (bottom third panel, Wilcoxon signed-rank test, $W = 27$, $p < .001$). Additional information on these analyses can be found in Appendix D.

Finally, motivated by the concept of boundary cells in neuroscience, which are activated when an animal is close to an environment’s boundary [47], we next looked for units which consistently encoded whether the agent was located in a border cell. We identified 373 units that were significantly correlated with this condition for all SIR types—far exceeding the number observed in a shuffled control (compare black and gray curves in the right-most bottom panel of Fig. 4, Wilcoxon signed-rank test, $W = 2$, $p < .001$). These invariant units were primarily concentrated in intermediate layers (approximately layers 8–18).

Overall, our findings indicate that the LLaMA-3.1-8B model builds an internal representation of grid space that is partially dependent on how spatial information is represented, but also contains components that show a high level of abstraction in encoding spatial features such as the location of the agent in 2D space.

6 Robust Units Encode Spatial Reasoning Across Unrelated Tasks

Spatial reasoning does not only involve representing locations in space—it requires detecting relations between objects’ positions to make predictions and guide decision-making. For an LLM to effectively navigate its environment, it must be able to understand whether its actions will move it closer to or farther from its target. This capability—predicting action correctness—requires an internal representation of the relative positions of objects within the environment. With this in mind, we next investigated whether the LLaMA-3.1-8B model exhibits this form of spatial reasoning representations.

We again fitted a linear model per parameter, this time using activations during the generation of the action keyword as predictors of action correctness (i.e., whether a selected action moves the agent closer to the goal) and applying a Bonferroni correction for the total number of parameters in the model. Our results revealed that all model layers contained hundreds to thousands of parameters that were significantly correlated with action correctness (Fig. 4, first panel), paralleling results from the spatial features studied in the previous section. Interestingly, both Cartesian representations—and especially the JSON variant—yielded the highest number of significant parameters, followed by Topographic and then Textual representations. The number and distribution of these parameters was quite consistent between SIRs of the same class (compare curves of the same

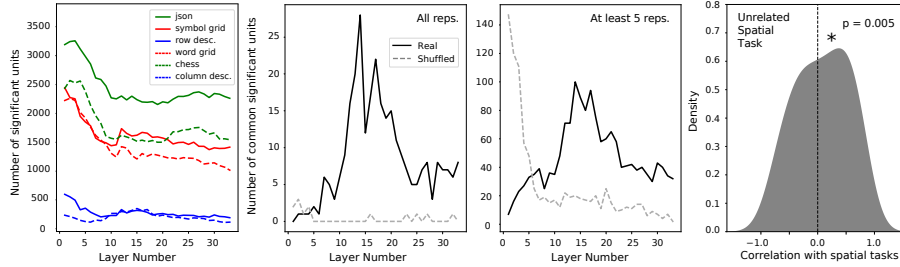


Fig. 4. Activations in LLaMA-3.1-8B predict action correctness across representation types. The left-most panel compares the total number of parameters per layer that are significantly correlated with action correctness for each SIR type. The second and third panels show the number of common significant parameters across all representation types and across at least 5 of the 6 representation types, respectively (black curves) compared to a shuffle between representations (gray dashed curves). The last panel shows the distribution of correlation coefficients between parameter activation and the spatial nature of a question in an unrelated task, for each parameter identified in the second panel as significantly correlated with action correctness for all SIR types in the GWSOT.

color in first panel), reinforcing the fact that differences between SIRs arise from the spatial encoding method rather than from specific prompt formatting. Interestingly, the parameter count for the JSON SIR was slightly higher than for the Chess Notation SIR, mirroring prior results from model’s behavior (Fig. 2, compare green curves in the first three panels—success rate, efficiency and final distance ratio tend to be improved for JSON with respect to Chess Notation). This suggests that prompt formatting could be playing a smaller but significant role in the Cartesian case.

We then asked if the model contained units that showed this correlation with action correctness irrespective of how spatial information was represented. Indeed, we identified a subset of 286 units that were significantly correlated with action correctness for all six SIR types (Fig. 4, second panel), far exceeding the 12 units found in a shuffled control (Wilcoxon signed-rank test, $W = 12.5$, $p < .001$). These 286 units were predominantly located in the middle layers of the model (approximately layers 12–22), consistent with the distribution of units encoding the agent’s position (Fig. 3) and with previous studies reporting that higher-level conceptual processing in LLMs tends to emerge in intermediate layers [26–29].

Given that this model had low performance in the GWSOT for some SIR types (i.e. both the Column Description and Word Grid SIRs had success rates below 20%), we considered that our inclusion criterion for this core group of parameters might be excluding units that nevertheless encode abstract spatial information. To explore this, we ran the same analysis using a more permissive threshold, identifying units that were significantly correlated with action correctness in at least five of the six SIR types. Under this relaxed criterion, we identified 1,560 units across all model layers (Fig. 4, third panel; there are more common units

than expected by chance: Wilcoxon signed-rank test, $W = 81.5$, $p < .001$). Notably, even with this substantial increase in the number of detected parameters, their distribution remained strikingly consistent: most of these units were concentrated in the middle layers, and almost none were located in earlier layers where most of the significant parameters for single SIRs are found.

These findings, together with our results from Figure 3, support the presence of a genuine spatial information encoding system within the model, characterized by a concentration of abstract, prompt-independent units in its middle layers. However, if these units are really functioning as a spatial world model, we would expect them to be involved in spatial reasoning regardless of the specific task. To test this, we examined the activation of the 286 core action-correctness encoding units in a different context. We presented the LLaMA-3.1-8B model with a set of spatial-reasoning and non-spatial-reasoning questions (200 each, examples in Appendix E) and averaged the activations of these core units across all input tokens. For each unit, we computed the correlation between its average activation and a binary indicator of whether the question was spatial or not. The resulting distribution of correlation coefficients was significantly higher than zero (Fig. 4, right-most panel, Wilcoxon signed-rank test, $W = 16634$, $p = 0.005$), indicating that these units are more active when receiving spatial-reasoning questions than when receiving questions that are not conveying information about space. In contrast, randomly chosen units—matching the number of core units per layer—did not exhibit this bias (Wilcoxon signed-rank test, $W = 19108$, $p = 0.31$). This suggests that these parameters, which predict action correctness in the GWSOT, are indeed processing spatial information in a task-independent way. Interestingly, when we performed the same analysis for the core units that encode the agent’s position in the 2D space, correlations were not significantly higher than 0 (Wilcoxon signed-rank test, $W = 129291$, $p = 0.78$), supporting the hypothesis that abstract spatial reasoning is related with inner representations of outcome prediction rather than with the encoding of specific spatial features such as object locations.

Finally, we asked if silencing these core spatial units would decrease the performance of the model when navigating our spatial orientation task. To evaluate this, we conducted an ablation experiment in which we silenced these 286 neurons during the 5×5 GWSOT using the JSON SIR (more information on ablation experiments can be found in Appendices B and C). Interestingly, there was almost no impact in the performance of the model: it achieved 55% success rate during ablation trials compared to 59% for the intact model. The same was true when silencing units correlated with the agent’s position for all SIR types (those found in Fig. 3): performance in this task was not reduced (62%). This suggests that abstract spatial units constitute only a minor component of the overall spatial encoding system in LLMs, and that SIR-specific units are enough to solve this task. To evaluate if SIR-specific units were indeed necessary, we attempted to test the model while ablating all units significantly correlated with action correctness for the JSON SIR. However, these comprised 59.7% of the total parameter count across all layers (see green curve in left-most panel of